

Unit 3 :: Complex Analysis

☞ **PHY0400404: Mathematical Physics** ☞

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Syllabus

Unit 3: Complex analysis

Functions of complex variables, analyticity and Cauchy-Riemann conditions, examples of analytic functions; singular functions: poles and branch points, order of singularity; integration of functions with complex variable, Cauchy's integral theorem and Cauchy's integral formula, simply and multiply connected regions; Laurent and Taylor's series expansions, residue theorem with application.

1 Functions of complex variables

2 Complex analytic functions

2.1 Cauchy-Riemann conditions

2.2 Examples of analytic functions

3 Singularities of complex functions

3.1 Poles

3.2 Branch points

4 Complex integration

4.1 Simply and multiply connected regions

4.2 Cauchy's integral theorem

4.3 Cauchy's integral formula

5 Complex series

5.1 Taylor's series expansion of a complex function

5.2 Laurent's series expansion of a complex function

5.2.1 Residue theorem